Heat-engine efficiency

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Calculate useful output and rejected energy for a heat engine.

SUCCESS CRITERIA - I can...

1. Use efficiency=work output/heat input.
2. Account for rejected heat.
3. Distinguish an ideal upper bound from actual performance.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “efficiency” and “cycle” to explain your answer. Sentence starters are optional.

efficiency: useful output divided by input

cycle: a process that returns a system to its initial state

kelvin: the absolute temperature scale used in thermal ratios

01 Warm-up: bring back an earlier idea

At the same kelvin temperature, sample B contains **3** times as many particles of the same ideal gas as A. **Compare average energy per particle** and total translational energy.

Word help: temperature = a measure of thermal state; it is not an energy amount

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $\eta = W_{out}/Q_{hot}$; $Q_{hot} = W_{out} + Q_{cold}$
$\eta_{max} = \mathbf{1} - T_{cold}/T_{hot}$.
Cyclic heat engine; heat amounts are positive magnitudes. The reversible upper bound uses kelvin temperatures.

Worked reasoning: A heat engine takes **100 J** from a hot source, does **25 J** of work and rejects **75 J**. Efficiency=**25/100=25%**.

Your turn: Draw three labeled energy arrows and **check conservation**.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .

Draw and label all energy transfers across the boundary.

Chosen system

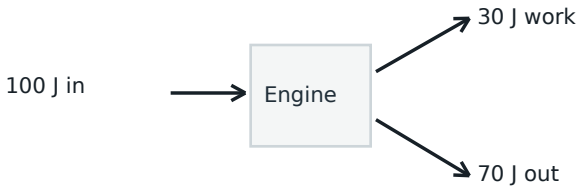
Stored energy change: _____ J

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

An engine receives **100 J**, produces **30 J** of work and rejects **70 J**. **Check the energy balance** and **find efficiency**.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: ① ② ③

04 Apply the physics

A heat engine receives **200 J** from a hot source per cycle and produces **50 J** of useful work. **Complete the energy-flow diagram**. **Find efficiency** and the heat rejected each cycle.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

Why should a **95%** efficient electric motor not be confused with a **95%** heat engine converting heat from one source?

Sentence starter: I think … . I can point to …, which shows …, so … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

At fixed mass, speed becomes **1/4** of its original value.
Compare kinetic energy and the energy-equivalent TNT amount.
Show the energy changes and transfers; **label units**.

Word help: kinetic energy = energy associated with motion

Sentence starter: Both … . They differ in … because … .

Initial	Transfers	Final

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

07 Exit check

A device outputs **40 J** work from **100 J** heat.
1. **State efficiency** and rejected heat.
2. **Label energy entering**, leaving or stored on the diagram of the objects you are studying.

Sentence starter: In my work, I used … to show … . I labeled it … .

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

An engine report lists **100 J** of input, **40 J** of useful work, and **50 J** of rejected heat.
1. **Find what** is missing from this energy account. **Give one possible explanation**.
2. On the diagram, label energy entering, leaving, or staying in the objects you are studying.

Sentence starter: My design includes … . It does what is needed by … .

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3